INCH-POUND
MIL-PRF-3098/120E
11 July 2011
SUPERSEDING
MIL-PRF-3098/120D
27 July 2005

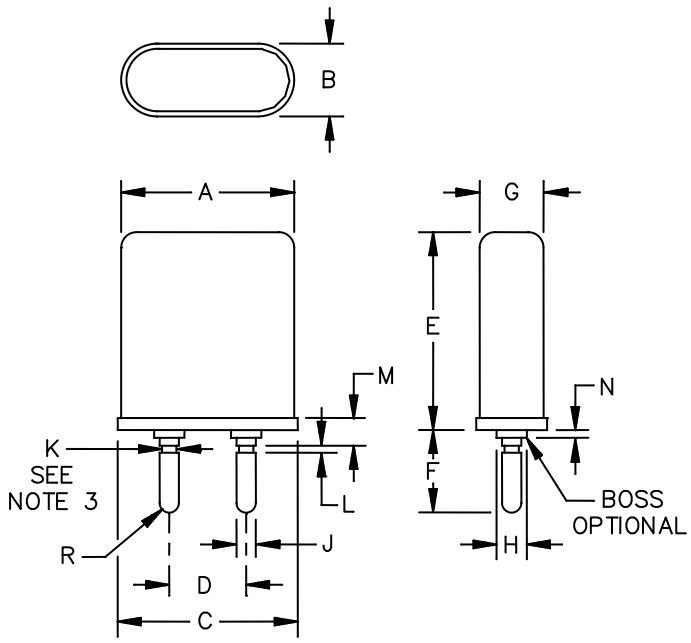
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR131/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; controlled; antiresonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.725	---	18.42
B	---	.352	---	8.94
C	---	.757	---	19.23
D	.460	.476	11.68	12.09
E	---	.775	---	19.69
F	.223	.248	5.66	6.30
G	---	.317	---	8.05
H	.075	.141	1.91	3.58
J	.048	.052	1.22	1.32
K	.030	.037	0.76	0.94
L	.015	.025	0.38	0.64
M	.010	.070	0.25	1.78
N	.015	.025	0.38	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR131/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Frequency stability: ±5 ppm.

Antiresonance, load capacitance: 30.0 pF ±0.5 pF.

Frequency and equivalent resistance: See table I.

Reference temperature: 75°C ±1°C.

Frequency tolerance: ±20 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	630
0.85+ to 0.90	600
0.90+ to 1.00	580
1.00+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.50	410
1.50+ to 1.62	380
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2.00	290
2.00+ to 2.12	270
2.12+ to 2.25	250
2.25+ to 2.60	200
2.60+ to 3.00	150
3.00+ to 3.40	110
3.40+ to 3.75	90
3.75+ to 4.00	75
4.00+ to 5.00	60
5.00+ to 7.00	35
7.00+ to 10.00	24
10.00+ to 15.00	22
15.00+ to 20.00	20

Mode of oscillation: Fundamental.

Operable temperature range: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating temperature range (controlled): +70°C to +80°C, inclusive.

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Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse): The violation of $P = I^2R$ is intentional at the higher resistance of the crystal unit.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz ±15 percent.
2.0 MHz + to 20.0 MHz ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz ±15 percent.
2.0+ MHz to 20.0 MHz ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: This crystal unit is identical to the CR27/U, except for a load capacitance rating of 30 pF ±0.5 pF. This unit is not interchangeable with the CR27/U.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR131/U or CR131B/U (followed by specified frequency). For Product Level S, PIN shall be CR131S/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-PRF-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR

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